

6.8 Quiz: Polynomials — Markscheme

1. [Maximum mark: 8]

Let $f(x) = x^2 - 4x - 12$.

(a) Write $f(x)$ in factored form $(x - a)(x - b)$. [2]

$$f(x) = (x - 6)(x - (-2)) = (x - 6)(x + 2) \quad \mathbf{A1A1}$$

Accept $(x + 2)(x - 6)$.

(b) Write down the x -intercepts. [1]

$$x = 6 \text{ and } x = -2 \quad \mathbf{A1}$$

Accept $(6, 0)$ and $(-2, 0)$.

FT: award A1 for x -intercepts read from the candidate's factors in (a).

(c) Write down the y -intercept. [1]

$$f(0) = -12, \text{ so the } y\text{-intercept is } (0, -12). \quad \mathbf{A1}$$

Accept -12 .

FT: award A1 for y -intercept correctly computed from the candidate's factored form in (a).

(d) Write $f(x)$ in vertex form $(x - h)^2 + k$. [3]

$$\text{Axis of symmetry: } x = \frac{-2 + 6}{2} = 2 \quad \mathbf{M1}$$

$$f(x) = (x - 2)^2 - 16 \quad \mathbf{A1A1}$$

Accept $h = 2, k = -16$.

M1 may also be awarded for any attempt at completing the square.

(e) Write down the coordinates of the vertex. [1]

$$\text{Vertex: } (2, -16) \quad \mathbf{A1}$$

FT: award A1 for vertex coordinates read from the candidate's vertex form in (d).

2. [Maximum mark: 7]

Consider the function $g(x) = -(x - 3)^2 + 16$.

(a) Write down the coordinates of the vertex. [2]

$$\text{Vertex: } (3, 16) \quad \mathbf{A1A1}$$

(b) Write down the equation of the axis of symmetry. [1]

$$x = 3 \quad \mathbf{A1}$$

(c) Show that $g(x) = -x^2 + 6x + 7$. [2]

$$g(x) = -(x - 3)^2 + 16 = -(x^2 - 6x + 9) + 16 \quad \mathbf{M1}$$

$$= -x^2 + 6x - 9 + 16 = -x^2 + 6x + 7 \quad \mathbf{A1 \ AG}$$

(d) Solve $g(x) = 0$ to find the x -intercepts. [2]

$$-x^2 + 6x + 7 = 0$$

$$x^2 - 6x - 7 = 0$$

$$(x - 7)(x + 1) = 0$$

$$x = 7 \text{ or } x = -1 \quad \mathbf{A1A1}$$

3. [Maximum mark: 6]

The line $y = 3x$ and the parabola $y = x^2 - 4$.

(a) Show that the x -coordinates of intersection satisfy $x^2 - 3x - 4 = 0$. [1]

At intersection: $x^2 - 4 = 3x$ **M1**

$x^2 - 3x - 4 = 0$ **AG**

(b) Show that $x^2 - 3x - 4 = 0$ has two distinct real roots. [1]

$\Delta = (-3)^2 - 4(1)(-4) = 9 + 16 = 25 > 0$ **M1**

Since $\Delta > 0$, there are two distinct real roots.

(c) Factorise $x^2 - 3x - 4$. [1]

$(x - 4)(x + 1)$ **A1**

(d) Find the x -coordinates of the points of intersection. [1]

$x = 4$ and $x = -1$ **A1**

FT: award A1 for x -coordinates read from the candidate's factors in (c).

(e) Find the coordinates of both points of intersection. [2]

$x = 4$: $y = 3(4) = 12$, so $(4, 12)$ **A1**

$x = -1$: $y = 3(-1) = -3$, so $(-1, -3)$ **A1**

FT: award A1 for each y -value correctly computed from the candidate's x -value.

4. [Maximum mark: 7]

The equation $3x^2 - kx + 12 = 0$ has two equal roots.

(a) Write down the discriminant in terms of k . [1]

$\Delta = (-k)^2 - 4(3)(12) = k^2 - 144$ **M1**

(b) Find the values of k . [2]

Equal roots: $\Delta = 0$ **M1**

$$k^2 - 144 = 0$$

$$k^2 = 144$$

$$k = \pm 12$$
 A1

(c) For $k = 12$, find the repeated root. [2]

$$3x^2 - 12x + 12 = 0$$

$$x^2 - 4x + 4 = 0$$

M1

$$(x - 2)^2 = 0$$

$$x = 2$$

A1

(d) State the values of k for which the equation has no real roots.

[2]

$$\Delta < 0$$

M1

$$k^2 - 144 < 0$$

$$k^2 < 144$$

$$-12 < k < 12$$

A1

Accept $|k| < 12$.